

I am overweight, doctor

Measurements to quantify obesity

- Weight
- Body Mass Index (BMI) = wt / ht^2
- Waist-to-hip ratio
- Waist circumference
- Skin-fold thickness
- Estimates of total body fat

WHO Expert Consultation 2004

- Caucasian based BMI cut off values are not directly applicable to other ethnic groups
- Asian populations
 - BMI $< 18.5 \text{ kg/m}^2$ – underweight
 - BMI 18.5 to 23 kg/m^2 – acceptable risk
 - BMI 23 to 27.5 kg/m^2 – increased risk
 - BMI $> 27.5 \text{ kg/m}^2$ – high risk

BMI cut offs for classification of obesity for Asians

BMI category	Range of BMI values
Underweight	$< 18.5 \text{ kg/m}^2$
Normal	$\geq 18.5 \text{ and } < 23.0 \text{ kg/m}^2$
Overweight	$\geq 23.0 \text{ and } < 25.0 \text{ kg/m}^2$
Obese	$\geq 25.0 \text{ kg/m}^2$
Obese I	$\geq 25.0 \text{ and } < 30.0 \text{ kg/m}^2$
Obese II	$\geq 30.0 \text{ kg/m}^2$

Measurements to quantify obesity

- BMI does not provide any indication of the distribution of fat
- Adipose tissue in a truncal distribution (ie, around and in the abdomen) has a particularly strong relationship with the adverse metabolic and vascular effects of obesity
- Distribution of fat can be assessed clinically by waist: hip ratio (WHR) or waist circumference.
WHO: $WHR \geq 0.9$ in men, $WHR \geq 0.85$ in women

Waist circumference

- Increasingly used as a measure of central obesity
- Ethnic differences recognised (IDF 2005)
- Europids: ≥ 94 cm in men, ≥ 80 cm in women
- Chinese: ≥ 90 cm in men, ≥ 80 cm in women

HK Population Health Survey 2014/15

- Among persons aged 15-84 by BMI:
 - 30% obese, 20% overweight
- Male: 36% obese, 21% overweight
 - 28% central obesity by WC
- Female: 24% obese, 19% overweight
 - 37% central obesity by WC
- Analysed by household income, in general, the proportion of persons classified as overweight or obese decreased with increasing household income
- The prevalence of central obesity generally increases with age

Diseases and morbidity attributable to obesity

Cardiovascular

Hypertension,
Coronary heart disease, Stroke

Metabolic

Type 2 Diabetes mellitus
Dyslipidaemia
Insulin resistance

Polycystic ovary syndrome

Gastrointestinal

Hiatus hernia, gall-stones
Colorectal cancer
Non alcoholic hepatic steatosis

Respiratory

Breathlessness
Obstructive sleep apnoea
Hypoventilation

Diseases and morbidity attributable to obesity

Neurological

Nerve entrapment

Sciatica

Breast

Breast cancer

Gynaecomastia

Genitourinary

Stress incontinence

Reduced fertility (PCOS)

Pregnancy complications

Orthopaedic

Osteoarthritis of weight-bearing joints

Psychological

Poor self-esteem, depression

Aetiology of obesity

- Obesity is due to multiple influences including genetic, environmental and behavioural factors
- Mismatch between energy intake and energy expenditure
- Only very few cases of obesity is solely due to single-gene defect
- Genetic influences are generally polygenic
- Adoption studies and complex segregation analysis suggest heritability of about 33%

Aetiology of obesity

- Environmental factors
- obesity is associated with food intake and content of fat in diet
- sedentary lifestyle
- inverse relationship between socio-economic studies and the prevalence of obesity

Conditions associated with obesity

- Hypothyroidism
 - Cushing's syndrome
 - Hypothalamic tumours
 - Growth hormone deficiency.
 - Hypogonadism in male.
 - Polycystic ovary syndrome
- } associated with increase in abdominal and visceral fat

Common medications causing weight gain

- Antidepressants: monoamine oxidase inhibitors, tricyclic antidepressants (nortriptyline, amitriptyline, doxepin), paroxetine, citalopram, escitalopram, imipramine, mirtazapine
- Antipsychotics: thioridazine, olanzapine, risperidone, clozapine, quetiapine
- Diabetes medications: eg, insulin, sulfonylureas, thiazolidinediones, meglitinides
- Glucocorticoids: eg, prednisone
- Anticonvulsants: valproate, carbamazepine

Regulation of energy balance and adiposity

- Hypothalamus - a centre for regulation of energy balance
- Integrates neural, hormonal and nutrient messages from the body and send signals to higher centres leading to feeling of hunger or satiety
- Controls energy expenditure via autonomic nervous system and pituitary hormones
- A large number of hypothalamic neurotransmitters affect both food intake and thermogenesis.

Leptin

- Peptide hormone synthesised by adipocytes
- Acts in the hypothalamus to suppress food intake and increase energy expenditure
- Mutation in leptin gene → ob/ob mice
Mutation in leptin receptor gene → db/db mice, fa/fa rat
- Mutations in leptin or leptin receptor as a cause of obesity are extremely rare in humans

Leptin

- Leptin levels rise in parallel with fat mass
- Leptin concentration is increased in obese subjects
- Human obesity is not a state of leptin deficiency or of extreme leptin insensitivity
- ? Partial resistance

A new perspective on Adipose Tissue

- Adipose tissue is not an inert tissue
- A major endocrine and secretory organ
- Release a wide range of protein factors and signals termed adipokines (eg adiponectin) , in addition to fatty acids and other lipid moieties

Why is sustained weight loss difficult?

- Longterm signals of energy stores and short-term fluctuations in food intake are released from adipose tissues and the gut endocrine system.
- These signals are integrated in the hypothalamus and brainstem and modulate neuropeptides release, resulting in subsequent changes in appetite, behaviour and energy expenditure to maintain a stable weight.
- Physiologoical adaptations (eg. increase in appetite-stimulating hormones like ghrelin and decrease in appetite suppressing hormones) to defend against weight loss.

Health costs of obesity

- Obesity associated with many co-morbid conditions
- risk of morbidity and mortality increases with an increase in body weight
e.g risk of CHD doubled if $BMI > 25$ and quadrupled if $BMI \geq 29$
- As weight reduction can alleviate obesity-associated co-morbidities, the goal of obesity treatment should be in the reduction of co-morbidity rather than for cosmetic reasons

Management of obesity

Aims

- weight reduction
- maintenance of weight loss
- modification of concurrent risk factors for mortality and morbidity such as smoking, diabetes, hypertension and hyperlipidaemia
- BE REALISTIC. Treatment directed towards health rather than cosmetic goals
- Weight loss of 5-10% of total body weight is already associated with health benefit

Strategies for weight reduction

- Set a realistic goal
- Lifestyle modification - diet and exercise
- Drug therapy
- Surgery

Diet therapy

- Cornerstone of any treatment program for obesity
- LCD: ≥ 800 kcal/d, typically 800-1500 kcal/d
- VLCD: <800 kcal/d
- Wt loss of around 0.5 kg/week is optimal. (600 kcal/day deficit). Faster rates of wt loss will lose not only fat but lean body mass
- VLCD leads to a faster rate in weight reduction in first 2 - 3 months but not superior in maintenance of weight loss after 1 year. Should only be used under medical supervision
- Longterm weight loss in most trials (FU 2 - 7 years) show weight loss in the range of 2 to 6 kg

Diet therapy

- Effective in promoting significant weight loss (max 4-7%)
- Low carbohydrate diets are somewhat more effective in short terms (3-6 months)
- All diets appear equivalent over long term (1 year or longer)

Physical activity

- Combining dietary modification with adequate physical activity has been repeatedly shown to be important for initial weight loss and necessary for longterm weight maintenance
- Gradual increase physical activity until energy expenditure of 1000 to 1500 kcal/week

Drug treatment

- Only appropriate in individuals who are at medical risk from their level of obesity and have not responded to more traditional, conservative management
- Consider in those who have BMI > 30 and in those with BMI of 27 - 29 if they have co-morbidities
- Treatment should be discontinued if patient does not get medical benefit

Pharmacological Treatment

- Gastrointestinal lipase inhibitor
- ~~Rimonabant~~ (endocannabinoid receptor antagonist, withdrawn in 2008)
- ~~Sibutramine~~ (serotonergic agent, withdrawn in 2010)
- ~~Lorcaserin~~ (selective serotonin 2C receptor agonist withdrawn in 2020)
- Phentermine/Topiramate (Qsymia)
- Bupropion/Naltrexone (Contrave)
- Liraglutide (Saxenda)

Orlistat

- A gastrointestinal lipase inhibitor
- promotes weight loss by inhibiting gastrointestinal lipase and thus lowering absorption of dietary fat.
- It decreases dietary fat by 30% with a dose of 120 mg three times a day
- The systemic absorption is negligible and potential for systemic adverse effects is small
- Possible side effects: GI upset, steatorrhoea, decreased absorption of fat soluble vitamins

Phentermine/Topiramate (Qsymia)

- Approved by FDA July 2012. (Not yet available in HK)
- Phentermine – anorexigenic agent, acts via enhancement of norepinephrine release and possibly via the blockade of norepinephrine reuptake as well.
- Topiramate – a neurostabilizer for treatment of seizure disorders and migraine prophylaxis. Exact mechanism of action for weight loss not known.
- Amount of weight loss achieved with combination therapy is of a greater magnitude than what could be achieved with either agent alone
- Adverse events: teratogenicity, a slight increase in heart rate, psychiatric and cognitive adverse effects, and metabolic acidosis

Bupropion/Naltrexone (Contrave)

- Approved by FDA Sept 2014
- Bupropion (inhibitor of neuronal reuptake of dopamine and norepinephrine) is used as an antidepressant to treat depression and as an aid to smoking cessation treatment
- Naltrexone (an opioid antagonist) is used to treat alcohol and opioid dependence
- Reduces appetite and the cravings for food
- Side effects: nausea, constipation, headache, vomiting, dizziness, insomnia, dry mouth and diarrhea. Monitor for mood changes, suicidal thoughts or actions

Liraglutide (Saxenda)

- Glucagon-like peptide-1 receptor agonist used in the treatment of type 2 diabetes
- Enhances glucose-stimulated insulin secretion and reduces glucagon secretion, delays gastric emptying and increases satiety by central effects on the hypothalamus
- A higher dose formulation developed specifically for the treatment of obesity
- Liraglutide 3 mg/day approved by FDA in Dec 2014
- Subcutaneous injection
- Side effects: mainly gastrointestinal

Current Pharmacotherapy for Obesity

Agent	Mechanism of Action	Mean Weight Loss ^{[1]*}
Liraglutide	GLP-1 receptor agonist	5.8 kg (5.4%)
Naltrexone/ bupropion	Opioid antagonist/ reuptake inhibitor of dopamine and norepinephrine	4.8%
Orlistat	Pancreatic and gastric lipase inhibitor	2.9 to 3.4 kg (2.9% to 3.4%)
Phentermine/ topiramate	Norepinephrine-releasing agent/ GABA receptor modulator	6.6 kg (6.6%) (recommended dose) 8.6 kg (8.6%) (high dose)

Novel therapy – Semaglutide (Wegovy)

- GLP-1 analogue, 94% homology to human GLP-1
- Half-life approx. 1 week and can be given as a weekly injection
- Approved for treatment of Type 2 DM at a dosage of 1 mg weekly
- High dose of 2.4 mg weekly approved by FDA for treatment of obesity in June 2021
- Mean weight loss approx. 15%
- Side effects: mainly gastrointestinal

Surgery

Patients considered eligible for surgery if

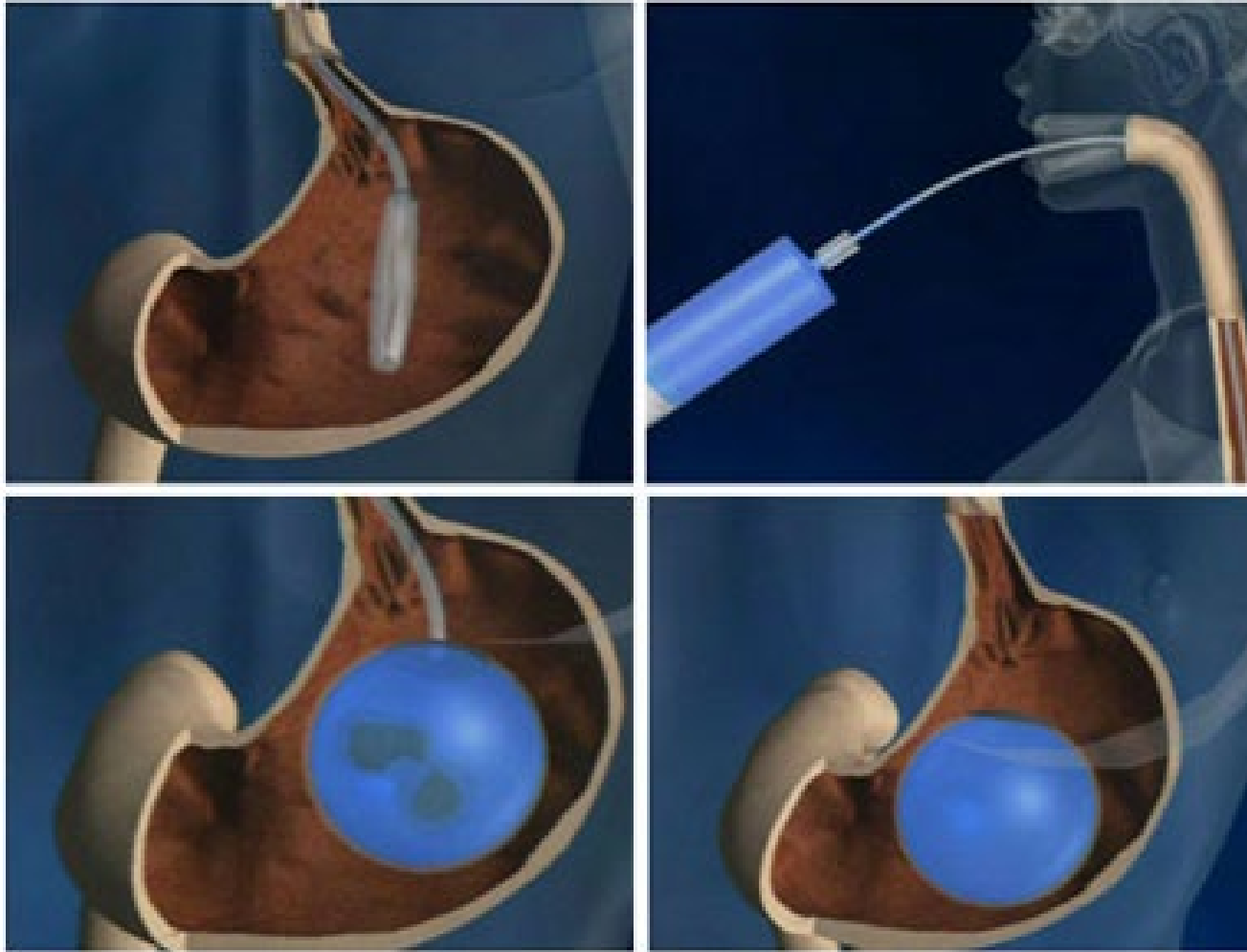
- BMI of $\geq 40 \text{ kg/m}^2$ without comorbidity
- BMI or $\geq 35 \text{ kg/m}^2$ with comorbidity (e.g. diabetes, respiratory insufficiency)

Surgery

Principles

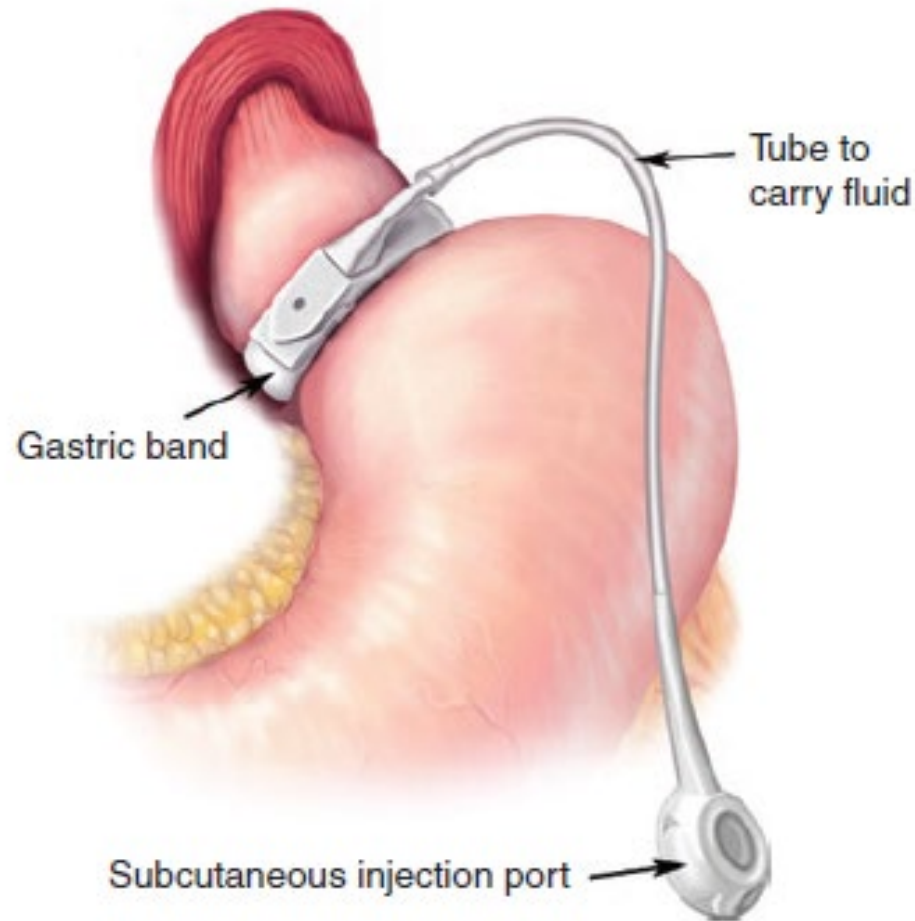
- Reduce the size of reservoir / intake
- Reduce the efficacy of absorption

Bioenterics intragastric balloon



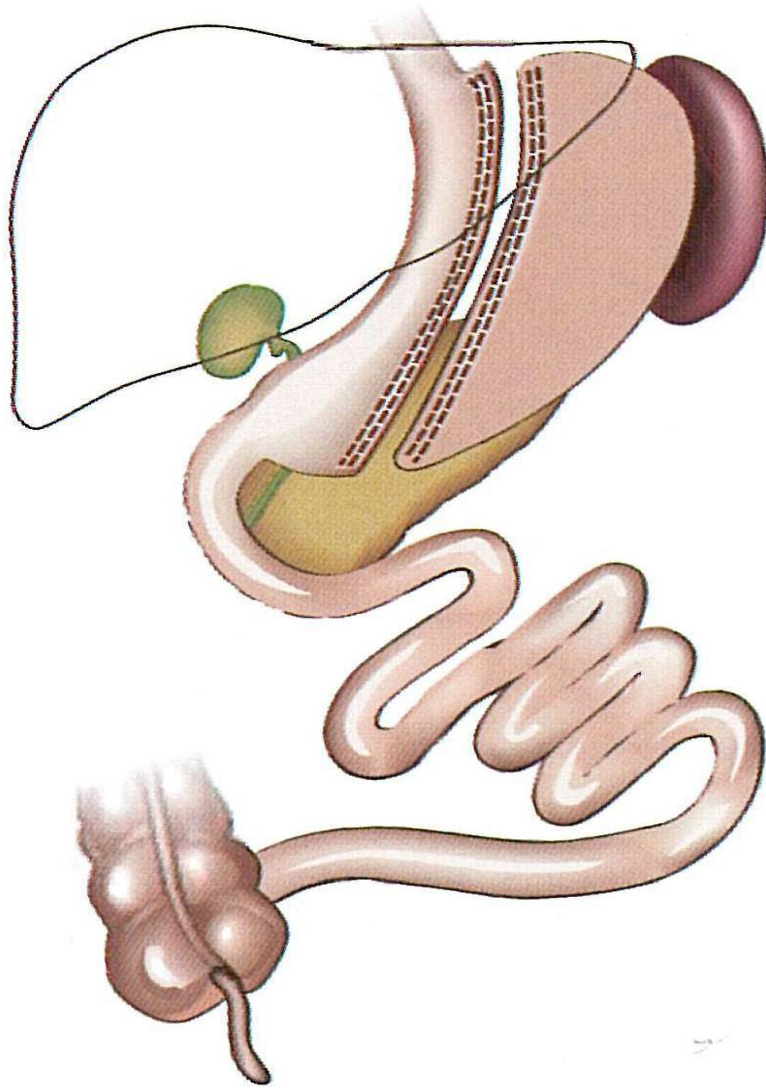
- Non-surgical procedure
- Placed inside stomach by endoscope and acts as a bezoar partially filling the stomach to induce early satiety.
- Non-permanent and reversible, remove after 6-9 months
- A kind of behavioral therapy

Laparoscopic adjustable gastric banding



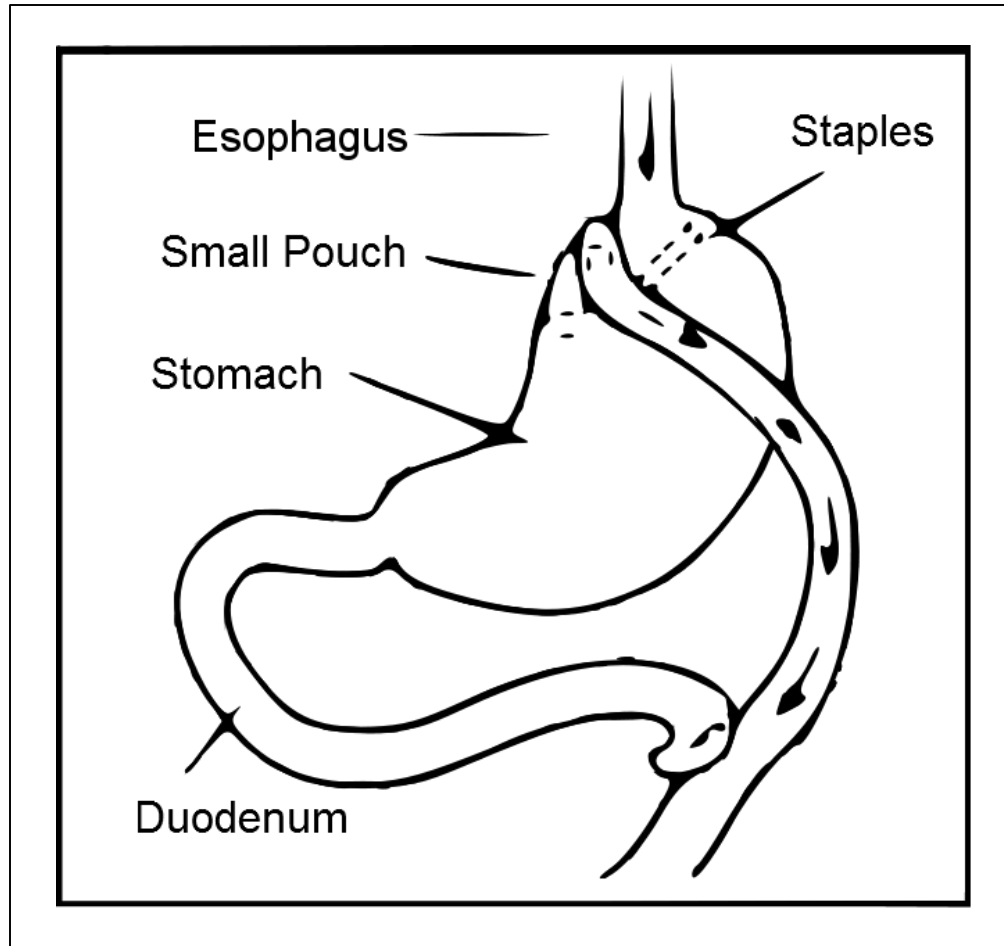
- A small inflatable belt placed around the upper portions of patient's stomach that functions to restrict the amount of food consumed and provides a constant feeling of being "full".
- Least invasive form of bariatric surgery

Laparoscopic Sleeve Gastrectomy



- Also known as vertical gastrectomy
- Greater curve of the stomach is excised by staples to create a new 150 cc banana-shaped gastric pouch.
- Restrict amount of food that can be eaten without any bypass of the intestines or malabsorption

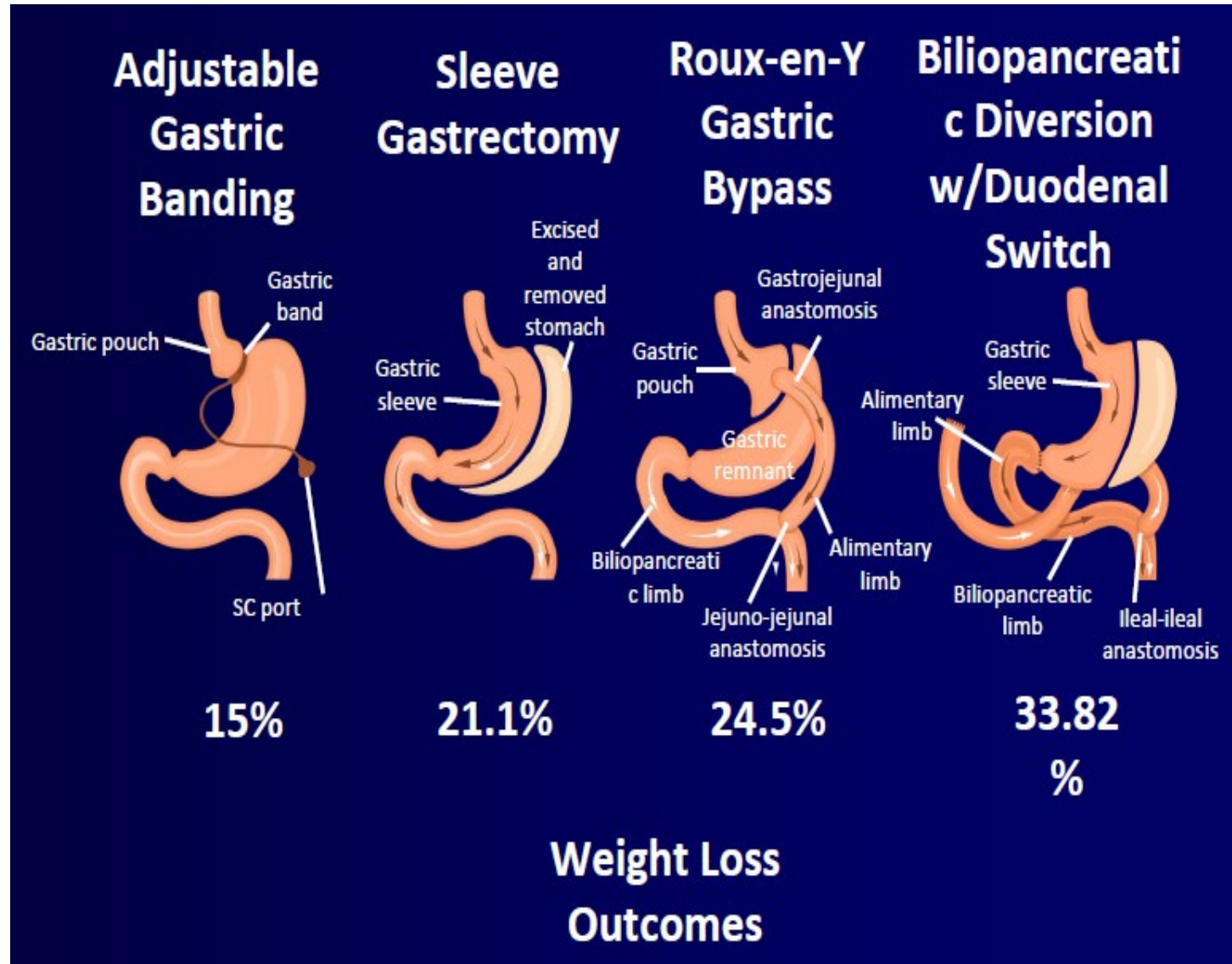
Roux-en-Y gastric bypass



Formation of a small gastric pouch (proximal portion of gastric body), a Roux-en-Y anastomosis to allow gastric, pancreatic, biliary and duodenal secretions to enter the distal intestine.

Bariatric Surgery

- Adjustable gastric banding (AGB) had lower mortality and complication rates. But reoperation rate was higher and weight loss was less substantial than GB.
- Gastric bypass (GB) was more effective in weight loss but associated with more complications.
- Sleeve gastrectomy more effective in weight loss than AGB and comparable to GB.



Complications of bariatric surgery

Longitudinal Assessment of Bariatric Surgery Consortium

- Adverse intraoperative events ~ 5% (10 clinical sites in US)

Longterm Complications	Frequency, %, and outcomes
Late complications for AGB	Surgical revision required in as many as 20 within 5 years
Band slippage	15
Leakage	2–5
Erosion	1–2
Late complications of bypass procedures	
Anastomotic strictures	1–5
Marginal ulcers	1–5
Bowel obstructions	1–2
Micronutrient and macronutrient deficiencies from RYGB 2–3 years after surgery (105)	
Iron deficiency	45–52
Vitamin B ₁₂ deficiency	8–37
Calcium deficiency	10
Vitamin D deficiency	51
Fat-soluble vitamin deficiencies (A, D, E, K) and protein calorie malnutrition from BPD and DS procedures	1–5
AGB, adjustable gastric banding; BPD, biliopancreatic diversion; DS, duodenal switch; RYGB, Roux-en-Y gastric bypass.	

Bariatric Surgery

- Mean 20 years weight reduction in the large Swedish Obese Subjects' study was 25% for Roux-en-Y gastric bypass and 15% for adjustable gastric banding
- An effective means to achieving lasting weight loss and improving metabolic disease and reducing cardiovascular risk in selected candidates
- Pre-op multi-disciplinary assessment is needed to select the appropriate candidates
- Choice of surgical modality should take the individual goals and existing co-morbidities into account
- Long-term follow-up is mandatory to support a safe outcome